PACKAGE LEAFLET

Directions for Use

Please Read Carefully!!!

hameln

MORPHINE-HAMELN 10 MG/ML INJECTION, 1 ML

CONTENT

One ampoule with 1.0 ml injection solution contains:

10 mg Morphine Sulphate 5 H2O equivalent to 7.52 mg of morphine.

DESCRIPTION

Clear and colourless to slightly yellow solution.

Manufacturer

Saneca Pharmaceuticals a.s.
Nitrianska 100
920 27 Hlohovec
Slovak Republic

INDICATION

Analgesic for severe and very severe pain.

RECOMMENDED DOSAGE AND METHOD OF ADMINISTRATION

Appropriate starting doses are:

Administration	*Adults*	*Children*
Intravenous	2.5-10 mg	0.05-0.1 mg/kg bodyweight
Subcutaneous, intramuscular	5-30 mg	0.05-0.2 mg/kg bodyweight
Epidural	1-4 mg	0.05-0.1 mg/kg bodyweight
Intrathecal	0.1-1 mg	0.02 mg/kg bodyweight

Adults

Intramuscular or subcutaneous

5-30 mg morphine sulphate. 10 mg is frequently used as a starting dose. This dose may, if necessary, be repeated every 4-6 h.

Intravenous

Only when a particularly rapid onset of effect is necessary. Slowly inject 2.5-10 mg morphine sulphate (10 mg per minute, if necessary dilution with isotonic sodium chloride solution).

Epidural

1-4 mg morphine sulphate (diluted with 10-15 ml isotonic sodium chloride solution).

Intrathecal

0.1-1.0 mg morphine sulphate (diluted with 1-4 ml isotonic sodium chloride solution).

Children

Intramuscular or subcutaneous

0.05-0.2 mg morphine sulphate/kg bodyweight; the single dose should not exceed 15 mg.

Intravenous

Only when a particularly rapid onset of effect is necessary. 0.05-0.1 mg morphine sulphate/kg bodyweight (dilution with isotonic sodium chloride solution is advisable).

Epidural

0.05-0.1 mg morphine sulphate/kg bodyweight (dilution with isotonic sodium chloride solution is advisable).

Intrathecal

0.02 mg morphine sulphate/kg bodyweight (dilution with isotonic sodium chloride solution is advisable).

The single doses during intramuscular, subcutaneous and intravenous use may be repeated as the effect diminishes, generally every 4-6 hours. On account of the longer duration of effect with epidural and especially with intrathecal use, with these methods of administration the daily dose is frequently the same as the individual dose.

Dosage for elderly patients

Patients of advanced age (generally aged 75 years and above) and patients in poor general physical condition may react more sensitively to morphine. Care must therefore be taken to adjust the dose more carefully and/or choose longer intervals between doses. It may be necessary to switch to lower doses.

A sufficiently high dose must always be given, while at the same time aiming for the lowest analgesically effective dose.

Posology and method of administration

Morphine injections may be given subcutaneously, intramuscularly, intravenously, epidurally or intrathecally.

The duration of use is at the discretion of the physician, and depends on the duration and severity of the disease.

The dosage of **Morphine-hameln** must be adjusted according to the severity of the pain and individual patient response.

The recommended range of individual doses for adults and children is to be understood as a guide for the individually adjusted dosage.

Treatment goals and discontinuation

Before initiating treatment with **Morphine-hameln**, a treatment strategy including treatment duration and treatment goals, and a plan for end of the treatment, should be agreed together with the patient, in accordance with pain management guidelines. During treatment, there should be frequent contact between the physician and the patient to evaluate the need for continued treatment, consider discontinuation and to adjust dosages if needed. When a patient no longer requires therapy with **Morphine-hameln**, it may be advisable to taper the dose gradually to prevent symptoms of withdrawal. In absence of adequate pain control, the possibility of hyperalgesia, tolerance and progression of underlying disease should be considered (see section **'WARNING & PRECAUTIONS'**).

Duration of treatment

Morphine-hameln should not be used longer than necessary.

CONTRAINDICATIONS

- Hypersensitivity to the active substance or to any of the excipients
- Existing intestinal occlusion (ileus)
- Disturbances of consciousness (coma)
- Respiratory depression or insufficiency
- Obstructive airways disease
- Cerebral trauma
- Increased intracranial pressure
- Hypotension with hypovolaemia
- Prostatic hypertrophy with formation of residual urine
- Ureteral stenosis
- Biliary tract diseases
- Obstructive and inflammatory intestinal diseases
- Phaeochromocytoma
- Pancreatitis
- Myxoedema
- Convulsive disorders
- Acute alcoholism
- Renal failure
- Liver failure
- Concurrent administration of MAO inhibitors or within two weeks of discontinuation of their use.

WARNING & PRECAUTIONS

Dose of morphine has to be readjusted following cordotomy operation.

Morphine can produce drug dependence and therefore has potential for being abused. Psychological/physical dependence and tolerance may also develop with repeated administration of morphine. However, respiratory depression, addiction, rapid tolerance and euphoria can be avoided by carefully titrating dose against pain. Clinically significant tolerance to morphine is unusual in cancer patients with severe pain.

For patients who no longer require morphine for relief of pain, doses should be gradually reduced in order to prevent withdrawal symptoms.

Risks from concomitant use with Benzodiazepines

Profound sedation, respiratory depression, coma, and death may result from the concomitant use of **Morphine-hameln** with benzodiazepines. Observational studies have demonstrated that concomitant use of opioids and benzodiazepines increases the risk of drug-related mortality compared to use of opioids alone. Because of these risks, reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate.

If the decision is made to newly prescribe a benzodiazepine and an opioid together, prescribe the lowest effective dosages and minimum durations of concomitant use.

If the decision is made to prescribe a benzodiazepine in a patient already receiving an opioid, prescribe a lower initial dose of the benzodiazepine than indicated in the absence of an opioid, and titrate based on clinical response.

If the decision is made to prescribe an opioid in a patient already taking benzodiazepine, prescribe a lower initial dose of the opioid, and titrate based on clinical response.

Follow patients closely for signs and symptoms of respiratory depression and sedation. Advise both patients and caregivers about the risks of respiratory depression and sedation when **Morphine-hameln** is used with benzodiazepines. Advise patients not to drive or operate heavy machinery until the effects of concomitant use of the benzodiazepine have been determined. Screen patients for risk of substance use disorders, including opioid abuse and misuse, and warn them of the risk for overdose and death associated with the use of benzodiazepines.

Serotonin Syndrome with Concomitant Use of Serotonergic Drugs

Cases of serotonin syndrome, a potentially life-threatening condition, have been reported during concurrent use of **Morphine-hameln** with serotonergic drugs. This may occur within the recommended dosage range.

Serotonin syndrome symptoms may include mental-status changes (e.g. agitation, hallucinations, coma), autonomic instability (e.g. tachycardia, labile blood pressure, hyperthermia), neuromuscular aberrations (e.g. hyperreflexia, incoordination) and/or gastrointestinal symptoms (e.g. nausea, vomiting, diarrhoea) and can be fatal. The onset of symptoms generally occurs within several hours to a few days of concomitant use, but may occur later than that. Discontinue **Morphine-hameln** if serotonin syndrome is suspected.

Oral P2Y12 inhibitor antiplatelet therapy

Within the first day of concomitant P2Y12 inhibitor and morphine treatment, reduced efficacy of P2Y12 inhibitor treatment has been observed (see section **DRUG INTERACTIONS**).

Adrenal insufficiency

Cases of adrenal insufficiency have been reported with opioid use, more often following greater than one month of use. Presentation of adrenal insufficiency may include non-specific symptoms and signs including nausea, vomiting, decreased appetite, fatigue, weakness, dizziness and low blood pressure. If adrenal insufficiency is suspected, confirm the diagnosis with diagnostic testing as soon as possible. If adrenal insufficiency is diagnosed, treat with physiologic replacement dosing of corticosteroids. Wean the patient of the opioid to allow adrenal function to recover and continue corticosteroid treatment until adrenal function recovers. Other opioids may be tried as some cases reported use of a different opioid without recurrence of adrenal insufficiency. The information available does not identify any particular opioids as being more likely to be associated with adrenal insufficiency.

Sexual function/Reproduction

Long term use of opioids may be associated with decreased sex hormone levels and symptoms such as low libido, erectile dysfunction, or infertility.

Elderly patients

The dose of morphine should be reduced in elderly patients and titrated to provide optimal pain relief with minimal side effects. Morphine clearance decreases and half-life increase in older patients.

Impaired renal function

Caution should be exercised with the use of morphine in patients with renal dysfunction i.e. renal failure, because metabolites can accumulate causing increased therapeutic and adverse effects.

Impaired liver function

Caution should be exercised in the use of morphine in patients with impaired liver function (e.g. cirrhosis) as elimination is likely to be affected. In severe hepatic impairment, the parent drug may not be readily converted to metabolite. The dose therefore should be carefully titrated and dosing interval may have to be more frequent to provide optimal pain relief.

Effects on ability to drive and use machines

Patients should not drive or operate machinery. Morphine may impair the mental and/or physical abilities required for the performance of potentially hazardous tasks, such as driving a car or operating machinery. Morphine in combination with other narcotic analgesics, phenothiazines, sedative-hypnotics, and alcohol have additive depressant effects.

Sleep-related breathing disorders

Opioids can cause sleep-related breathing disorders including central sleep apnoea (CSA) and sleep-related hypoxemia. Opioid use increases the risk of CSA in a dose-dependent fashion. In patients who present with CSA, consider decreasing the total opioid dosage.

Severe cutaneous adverse reactions (SCARs)

Acute generalized exanthematous pustulosis (AGEP), which can be life-threatening or fatal, has been reported in association with morphine treatment. Most of these reactions occurred within the first 10 days of treatment. Patients should be informed about the signs and symptoms of AGEP and advised to seek medical care if they experience such symptoms.

If signs and symptoms suggestive of these skin reactions appear, morphine should be withdrawn and an alternative treatment considered.

Hepatobiliary disorders

Morphine may cause dysfunction and spasm of the sphincter of Oddi, thus raising intrabiliary pressure and increasing the risk of biliary tract symptoms and pancreatitis.

Opioid Use Disorder (abuse and dependence)

Tolerance and physical and/or psychological dependence may develop upon repeated administration of opioids such as **Morphine-hameln**.

Repeated use of **Morphine-hameln** can lead to Opioid Use Disorder (OUD). A higher dose and longer duration of opioid treatment can increase the risk of developing OUD. Abuse or intentional misuse of **Morphine-hameln** may result in overdose and/or death. The risk of developing OUD is increased in patients with a personal or a family history (parents or siblings) of substance use disorders (including alcohol use disorder), in current tobacco users or in patients with a personal history of other mental health disorders (eg. major depression, anxiety and personality disorders).

Before initiating treatment with **Morphine-hameln** and during the treatment, treatment goals and a discontinuation plan should be agreed with the patient (see section '**Posology and method of administration**'). Before and during treatment the patient should also be informed about the risks and signs of OUD. If these signs occur, patients should be advised to contact their physician.

Patients will require monitoring for signs of drug-seeking behaviour (e.g. too early requests for refills). This includes the review of concomitant opioids and psycho-active drugs (like benzodiazepines). For patients with signs and symptoms of OUD, consultation with an addiction specialist should be considered.

DRUG INTERACTIONS

CNS depressants, Anxiolytics, Hypnotics

Morphine should be used with caution in patients who are concurrently receiving other central nervous system depres sants including sedatives or hypnotics, general anaesthetics, phenothiazines, other tranquilisers muscle relaxants, antihypertensives, gabapentin or pregabalin and alcohol. Interactive effects resulting in respiratory depression, hypotension, profound sedation, or coma may result if these drugs are taken in combination with the usual doses of morphine.

Muscle relaxants

Morphine may enhance neuromuscular blocking action of skeletal muscle relaxants.

Mixed agonist/antagonist opioid analgesics

e.g. pentazocine, buprenorphine, butorphanol or nalbuphine, can reduce the analgesic effect of morphine by competitively blocking the receptor.

Monoamine oxidase inhibitors (MAOIs)

MAOIs intensify the effect of morphine and other opioid drugs. Severe and fatal events (e.g. anxiety, confusion and significant respiratory depression, sometimes leading to coma) from the co-administration of both drugs. Therefore, morphine should not be given to patients taking MAOIs or within 14 days of stopping such treatment.

Cimetidine

Cimetidine decreases the metabolism of morphine. Confusion and severe respiratory depression were reported after a haemodialysis patient had received both morphine and cimetidine.

Diuretics

Morphine reduced the efficacy of diuretics by inducing the release of antidiuretic hormones.

Alcohol

Enhanced sedative and hypotensive effect.

Anti-Arrhythmics

Delayed absorption of mexiletine.

Domperidone and metoclopramide

Antagonism or gastro-intestinal effects.

Dopaminergics

Hyperpyrexia and CNS toxicity reported with selegiline.

Serotonergic drugs

The concomitant use of opioids with other drugs that affected the serotonergic neurotransmitter system has resulted in serotonin syndrome. If concomitant use is warranted, carefully observe the patient, particularly during treatment initiation and dose adjustment. Discontinue **Morphine-hameln** if serotonin syndrome is suspected. Examples of serotonergic drugs are selective serotonin reuptake inhibitors (SSRIs), serotonin and norepinephrine reuptake inhibitors (SNRIs), tricyclic antidepressants (TCAs), triptans, 5-HT3 receptor antagonists, drugs that affect the serotonin neurotransmitter system (e.g. mirtazapine, trazodone, tramadol), monoamine oxidase (MAO) inhibitors (those intended to treat psychiatric disorders and also others, such as linezolid and intravenous methylene blue).

Benzodiazepines

Due to additive pharmacologic effect, the concomitant use of opioids with benzodiazepines increases the risk of respiratory depression, profound sedation, coma and death.

The concomitant use of opioids and benzodiazepines increases the risk of respiratory depression because of actions at different receptor sites in the central nervous system that control respiration. Opioids interact primarily at μ-receptors, and benzodiazepines interact at GABAA sites. When opioids and benzodiazepines are combined, the potential for benzodiazepines to significantly worsen opioid-related respiratory depression exists.

Reserve concomitant prescribing of these drugs for use in patients for whom alternative treatment options are inadequate (see section '**WARNING & PRECAUTIONS**').

Limit dosage and duration of concomitant use of benzodiazepines and opioids, and follow patients closely for respiratory depression and sedation.

Oral P2Y12 inhibitor antiplatelet therapy

A delayed and decreased exposure to oral P2Y12 inhibitor antiplatelet therapy has been observed in patients with acute coronary syndrome treated with morphine. This interaction may be related to reduced gastrointestinal motility and apply to other opioids. The clinical relevance is unknown, but data indicate the potential for reduced P2Y12 inhibitor efficacy in patients co-administered morphine and a P2Y12 inhibitor (see section '**WARNINGS & PRECAUTIONS**') . In patients with acute coronary syndrome, in whom morphine cannot be withheld and fast P2Y12 inhibition is deemed crucial, the use of a parenteral P2Y12 inhibitor may be considered.

PREGNANCY AND LACTATION

Pregnancy

There is no adequate human data on which to base an assessment of the possible risk of teratogenicity. However, there are risk of withdrawal symptoms in neonates following prolonged use of morphine during pregnancy.

Morphine should only be used during pregnancy if the benefit to the mother outweighs the risks to the child. Due to its mutagenic properties, morphine should only be used in men who are potent and women of childbearing age if effective contraception is used.

Labour/delivery

Morphine may prolong or shorten the duration of labour. Neonates, whose mothers have been given opioid analgesics during labour, should be monitored for signs of respiratory depression and, if necessary, treated with a specific opioid antagonist (Naloxone hydrochloride) to reverse narcotic-in-duced respiratory depression. After chronic morphine use by the mother, newborns may develop withdrawal symptoms.

Breast-feeding

Morphine is excreted in human milk and breast-feeding is not recommended while a mother is receiving morphine. Withdrawal syndrome was observed in breast-fed infants after maternal administration of morphine had been stopped.

SIDE EFFECTS

Frequency	Symptoms
Very Common	• Nausea • Vomiting • Respiratory depression • Constipation • Sedation • Drowsiness • Disorientation • Itching • Sweating • Hallucinations • Dysphoria • Euphoria • Tolerance and dependence (with long term treatment)
Common	Skin • Urticaria • Skin rash • Pain at injection site • Contact dermatitis CNS and nervous system • Headache • Dizziness • Agitation • Convulsions • Impairment of taste • Mood changes • Changes of cognitive and sensory abilities • Insomnia • Intracranial hypertension • Tremor Musculoskeletal system • Muscle spasm Eye • Miosis • Visual disturbances Gastrointestinal system • Dryness of mouth • Stomach pain • Hiccups • Diarrhoea • Abdominal cramps • Biliary colic Cardiovascular system • Flushing • Chills • Orthostatic hypotension • Bradycardia • Hypertension • Tachycardia • Heart failure • Pulmonary oedema Respiratory system • Laryngeal spasm • Bronchospasm • Cough attenuation Urogenital system • Urinary retention or hesitancy • Impotence General • Oedema • Hypothermia • Hyperthermia • Withdrawal syndrome
Uncommon	• Anaphylactic reaction after intravenous injection.
Not known	Skin and subcutaneous tissue disorders • Acute generalized exanthematous pustulosis (AGEP) Respiratory disorders • Central sleep apnoea syndrome Gastrointestinal disorders • Pancreatitis Hepatobiliary disorders • Spasm of sphincter of Oddi

Post-marketing Experience:

Serotonin syndrome (see section '**WARNING & PRECAUTIONS**')

Adrenal insufficiency (see section '**WARNING & PRECAUTIONS**')

Androgen deficiency:

Cases of androgen deficiency have occurred with chronic use of opioids. Chronic use of opioids may influence the hypothalamic-pituitary-gonadal axis, leading to androgen deficiency that may manifest as low libido, impotence, erectile dysfunction, amenorrhea, or infertility. The casual role of opioids in the clinical syndrome of hypogonadism is unknown because the various medical, physical, lifestyle, and psychological stressors that may influence gonadal hormone levels have not been adequately controlled for in studies conducted to date. Patients presenting with symptoms of androgen deficiency should undergo laboratory evaluation.

Infertility:

Chronic use of opioids may cause reduced fertility in females and males of reproductive potential. It is not known whether these effects on fertility are reversible.

Drug dependence

Repeated use of **Morphine-hameln** can lead to drug dependence, even at therapeutic doses. The risk of drug dependence may vary depending on a patient's individual risk factors, dosage, and duration of opioid treatment (see section **'WARNING & PRECAUTIONS'**).

SYMPTOMS AND TREATMENT OF OVERDOSE

Symptoms

Overdose with morphine is characterized by respiratory depression (a decrease in respiratory rate and/or tidal volume, Cheyne-Stokes respiration, cyanosis), pinpoint pupils, extreme somnolence progressing to stupor and coma, skeletal muscle flaccidity, cold and clammy skin and sometimes bradycardia and hypotension. In severe overdose, apnoea, circulatory collapse, cardiac arrest and death may occur.

Treatment

In unconscious patients with respiratory arrest:

- Ventilation, intubation and intravenous administration of 0.4 mg naloxone.
- Measures to protect against heat loss and for volume therapy as needed.

In cases of persistent respiratory insufficiency the single dose must be repeated 1–3 times at three-minute intervals until respiration rate has normalized and the patient reacts to pain stimuli. Close monitoring (for at least 24 hours) is necessary as the effect of the narcotic antagonist is shorter than that of morphine so there is a possibility of reoccurrence of respiratory insufficiency.

For children, the dose of narcotic antagonist, naloxone is 0.01 mg per kg bodyweight per single dose.

PHARMACODYNAMICS

Morphine is pharmacologically the most important alkaloid of opium. It is used primarily as an analgesic for severe pain. The analgesic effect of morphine is primarily due to an interaction with the OP3(μ)-receptor, one of three major classes of opioid receptors that can be distinguished in the central nervous system. The metabolite morphine-6-glucoronide also acts as an agonist on opioid-receptors and contributes significantly to the pharmacological effects of chronic morphine treatment.

Morphine causes respiratory depression by diminishing the response of the respiratory centres to CO2. The effect is mediated by the action on OP3-receptors and can be antagonized by naloxone.

Morphine stimulates the chemo-receptor trigger zone by action on dopamine-receptors and may cause nausea and vomiting.

Other effects: Morphine can induce euphoria and miosis. Hypotension may occur due to increased histamine release, especially in hypovolaemic patients.

PHARMACOKINETICS

Approximately 35% of morphine is bound to plasma proteins. Morphine is rapidly metabolised. Approximately 50% is converted to the major metabolite, the pharmacologically inactive morphine-3-glucuronide and 10-15% is converted to the active morphine-6-glucoronide. Further metabolites include a diglucuronide, normorphine and its glucuronides. Approximately 60% of morphine is excreted in the urine in 24 hours, of which less than 10% of the dose is excreted as unchanged drug.

LIST OF EXCIPIENTS

Sodium chloride
Hydrochloric acid
Water for injection
Nitrogen

PRESENTATION

Package of 5 ampoules with 1 ml each

Package of 10 ampoules with 1 ml each

INSTRUCTIONS FOR USE

Dilution

When dilution of the **Morphine-hameln** Injection is necessary, sodium chloride 0.9% is recommended as diluent.

Chemical and physical in-use stability has been demonstrated with dilutions of **Morphine-hameln** in sodium chloride 0.9% or dextrose 5% for 24 hours at 30°C, or 72 hours at 2 to 8°C).

From a microbiological point of view, the dilutions should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user.

For recommended dilutions see section **'RECOMMENDED DOSAGE AND METHOD OF ADMINISTRATION'**.

Incompatibilities

Morphine salts are sensitive to changes in pH and morphine is liable to be precipitated out of solution in an alkaline environment. Compounds incompatible with morphine salts include aminophylline and sodium salts of barbiturates and phenytoin. Other incompatibilities (sometimes attributed to particular formulations) have included aciclovir sodium, cefepime hydrochloride, chlorothiazide sodium, cloxacillin sodium, doxorubicin, floxacillin sodium, furosemide, gallium nitrate, heparin sodium, methicillin sodium, pethidine hydrochloride, promethazine hydrochloride, sargramostim and tetracyclines. Specialised references should be consulted for specific compatibility information.

Physicochemical incompatibility (formation of precipitates) has been demonstrated between solutions of morphine sulphate and 5-fluorouracil.

SPECIAL PRECAUTIONS FOR STORAGE

Do not store above 30°C.

Keep ampoules inside the outer carton to protect from light. Keep out of reach of children.

Jauhi daripada kanak-kanak.

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